

REVISED MODEL DECK

Dynamic Sports Business Decision Model

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A state-based management system that links roster decisions, demand, pricing, financing, injuries, expansion, and franchise value.

Note: numerical scenario outputs are intentionally left for rerun because the original tables did not reconcile.



What the revision fixes first

The paper was already directionally strong; the upgrade makes the model internally consistent and judge-proof.

01

Results reconciliation

Flag old scenario outputs that contradicted the intended mechanisms.

02

Owner objective

Optimize profit + franchise value, not isolated weekly rules.

03

Demand calibration

Normalize brand and quality so price does not max out mechanically.

04

Roster consistency

Use one value definition from formula to implementation.

05

Risk validation

Stress-test cash, injuries, demand, costs, rates, and sponsorship.

Core message: the final submission should be judged on policy logic + corrected rerun outputs, not polished but inconsistent numbers.

One objective aligns every decision

The owner's policy should maximize total franchise value, not just weekly revenue.

$$\max E[\sum \delta^t (\Pi_t - \lambda \cdot \text{Risk}_t) + \theta \cdot V_T]$$

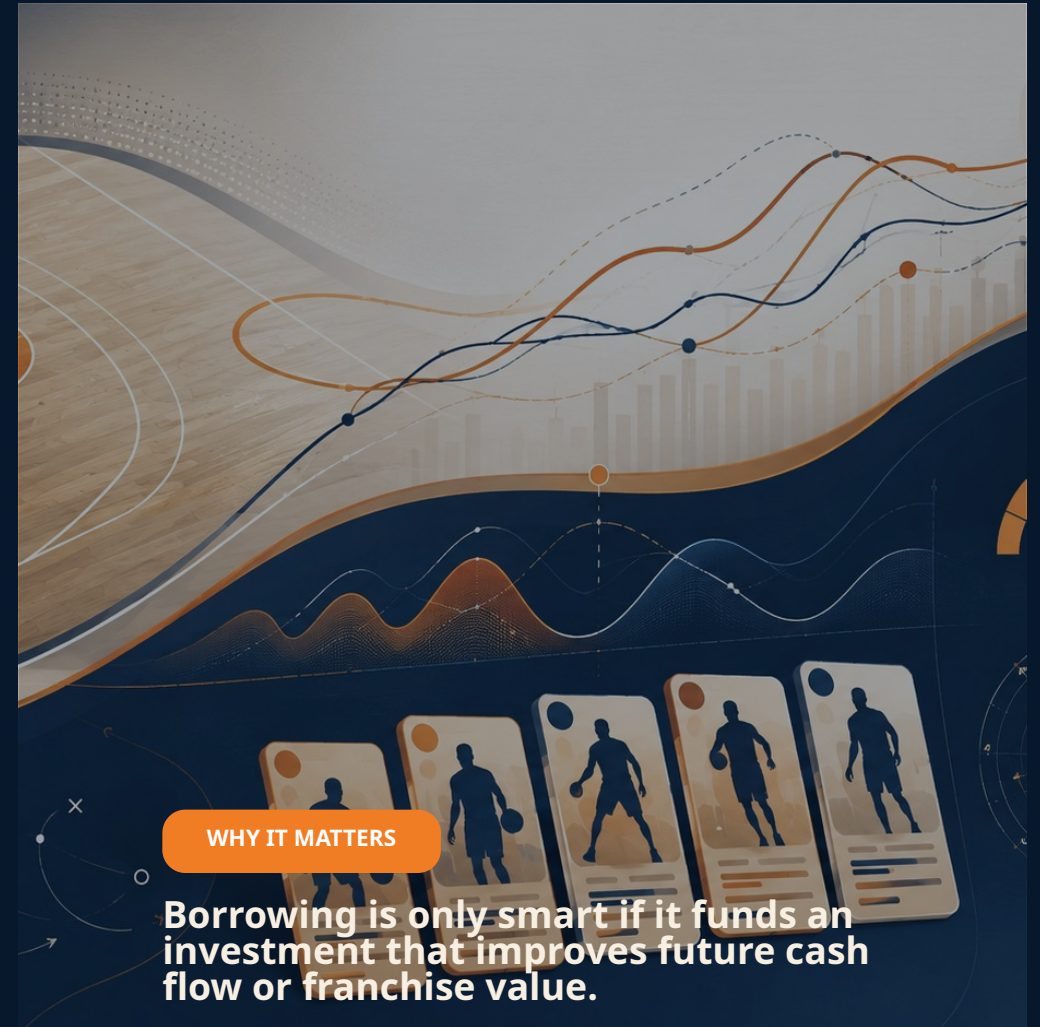
Where the policy π chooses debt, marketing, ticket prices, and roster actions from the current state:

$$X_t = (Q_t, B_t, C_t, D_t, M_t, E_t, V_t)$$

Π_t rewards short-run profit

Risk_t penalizes cash breaches, leverage, and competitive decline

V_T captures long-run franchise value from brand and revenue growth

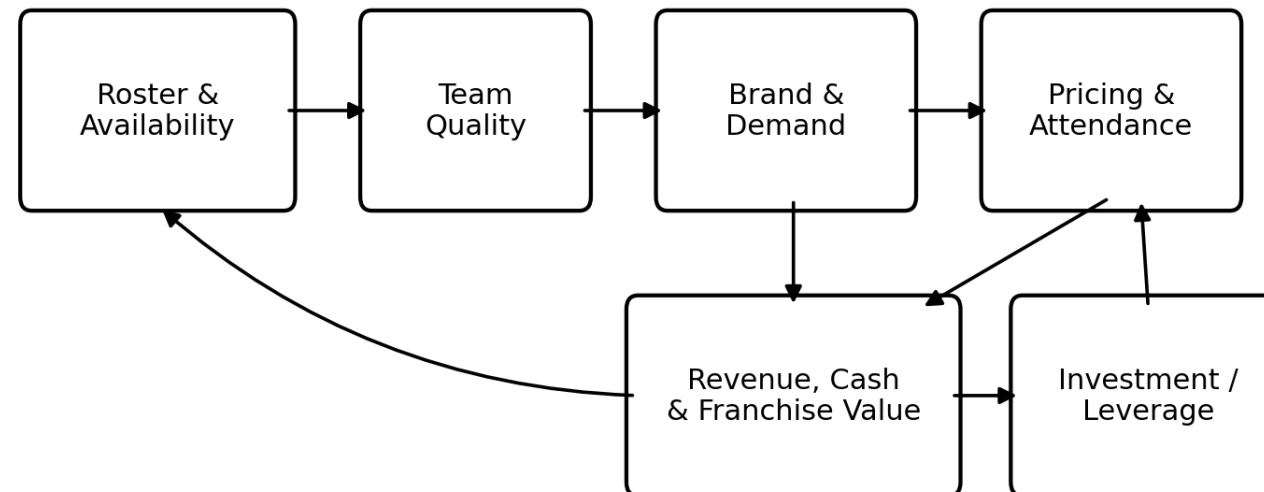


Dynamic feedback system

A true dynamic model specifies an evolving state and policies that update as the state changes.

Dynamic feedback system: actions alter future states

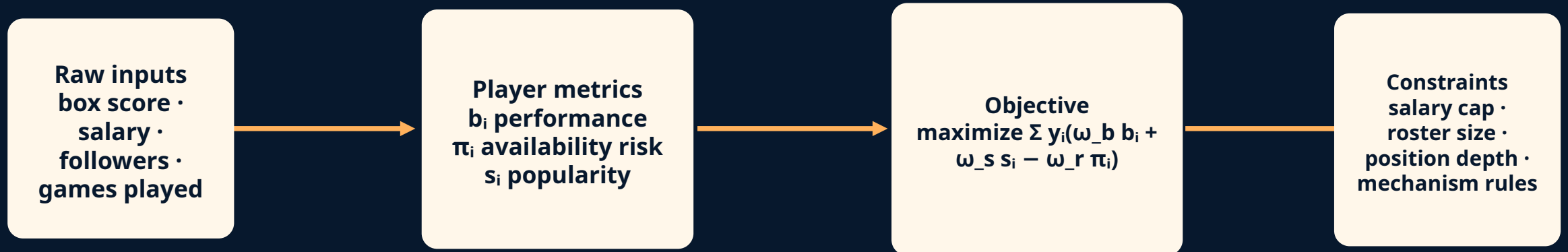
Performance and popularity create demand; demand creates cash; cash and financing constrain future roster and growth decisions.



Key correction: demand, cash, and franchise value are feedback variables — not one-time outputs.

Player data becomes a constrained roster decision

The revised paper keeps the model transparent while removing formula/implementation mismatch.



Implementation rule: exact binary optimization is preferred; a value-per-dollar heuristic is acceptable only if clearly labeled as an approximation.

Demand and pricing must be calibrated

The old output sent ticket price to the grid ceiling while attendance stayed near capacity.

Problem

Brand stock entered demand through an exponential term. If B keeps accumulating, the model can make almost any high price look optimal.

Revision

Normalize Q and B, cap demand realistically, and calibrate with observed attendance plus ticket-price anchors.

$$Att_t(p) = \min(Cap, A_0 \cdot \exp(\xi \tilde{B}_t + \zeta \tilde{Q}_t) \cdot p^{-\varepsilon})$$

$\tilde{B}$ and $\tilde{Q}$ are standardized indices, not raw accumulated stocks.

Expected check after rerun: price changes should trade off against attendance, not sit at pmax every week.

Leverage should fund positive-NPV actions

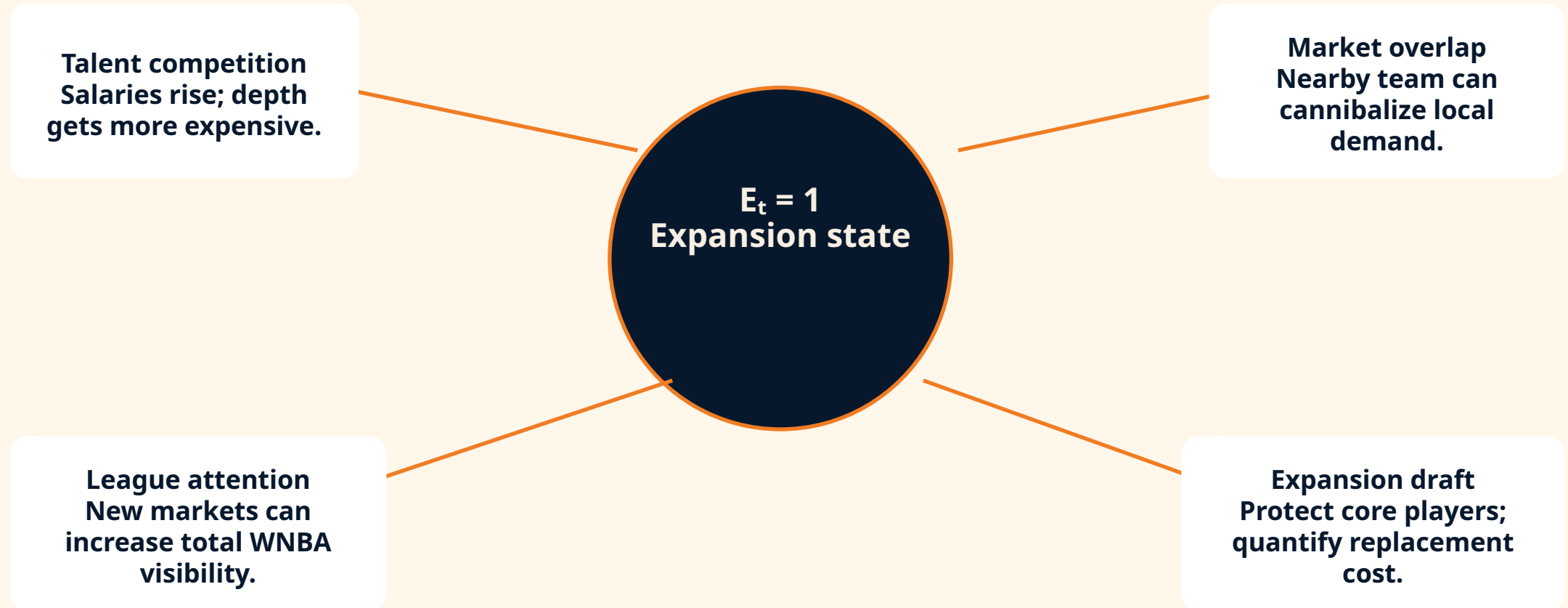
Debt is not a cash-padding variable; it must be tied to investments with expected returns.



Interest-rate correction: if r_0 is annual, convert to the weekly decision period before charging $r_t D_t$.

Expansion is a multi-channel shock

The revised model treats expansion as more than salary inflation.



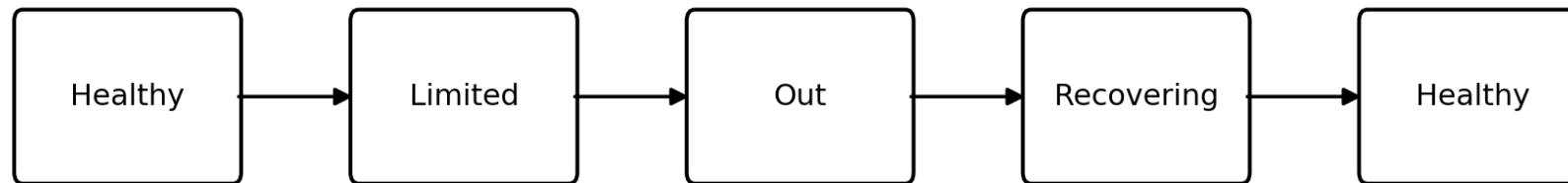
Recommended policy: higher cash buffer + value-per-dollar depth while the league is reallocating talent.

Injury response becomes persistent

A key-player injury should visibly move quality, pricing, roster decisions, and liquidity.

Injury-state process replaces a one-week mechanical shock

State-specific availability and performance multipliers feed directly into roster optimization, quality, pricing, and liquidity.

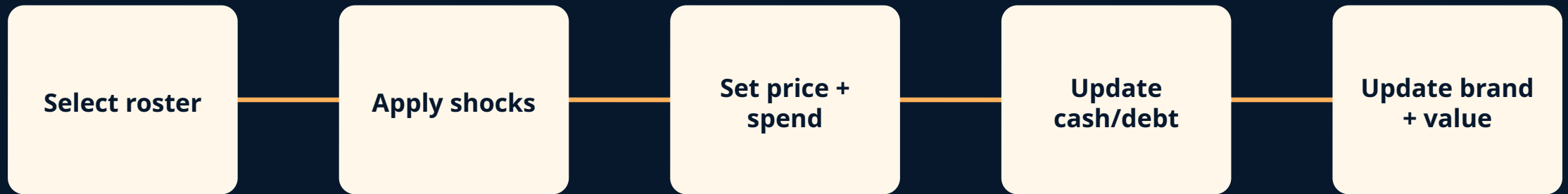


Example multipliers are parameters to calibrate, not medical probabilities.

Validation check: injury scenario must show $Q_t \downarrow$, waiver/free-agent option activation, price/demand adjustment, and cash/profit impact.

Simulation loop and validation gates

The revised paper separates mathematical design from numbers that still need to be rerun.



Validation gates before final outputs

deterministic baseline $\approx$ Monte Carlo mean under centered shocks
injury scenario changes quality and activates replacement logic
macro stress weakly lowers objective unless offset by explicit effects
attendance stays within realistic price-capacity behavior

Final report should show decisions, not raw dataframes

Once the model is rerun, replace the old screenshots with figures that tell the managerial story.

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|----|---------------------------------|--|
| 01 | Scenario profit + downside risk | Compare expected value and tail loss. |
| 02 | Price vs. attendance path | Show actual pricing tradeoff after calibration. |
| 03 | Injury response trajectory | Track injury state, Q_t , roster move, profit. |
| 04 | Cash and leverage paths | Expose liquidity stress and debt policy. |
| 05 | Monte Carlo profit distribution | Highlight 5% tail and CVaR. |

Keep the old raw tables in the appendix only after the corrected run is complete.

Recommendation to Owner + GM

Use the revised model as a contingent decision system: choose actions based on state, then update the state after every week.

Protect liquidity when macro or injury states worsen.

Invest only when debt funds measurable future demand or value.

Rerun all scenarios before reporting numerical conclusions.

Bottom line: stronger model first, numbers second.

